



1. Description

1.1. Project

Project Name	CUBEMX
Board Name	custom
Generated with:	STM32CubeMX 6.0.0
Date	12/27/2025

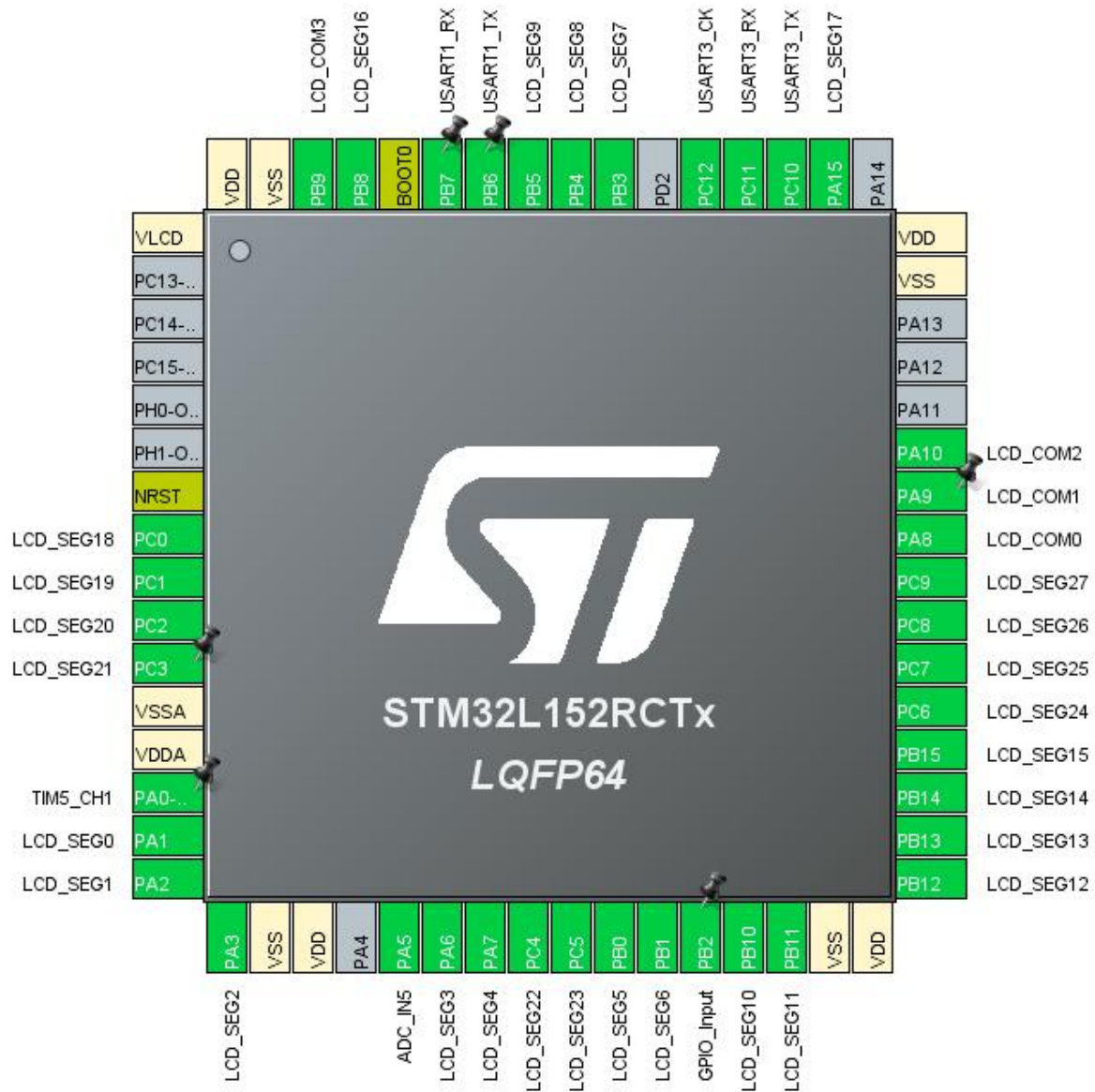
1.2. MCU

MCU Series	STM32L1
MCU Line	STM32L151/152
MCU name	STM32L152RCTx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	Arm Cortex-M3
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2. Pinout Configuration



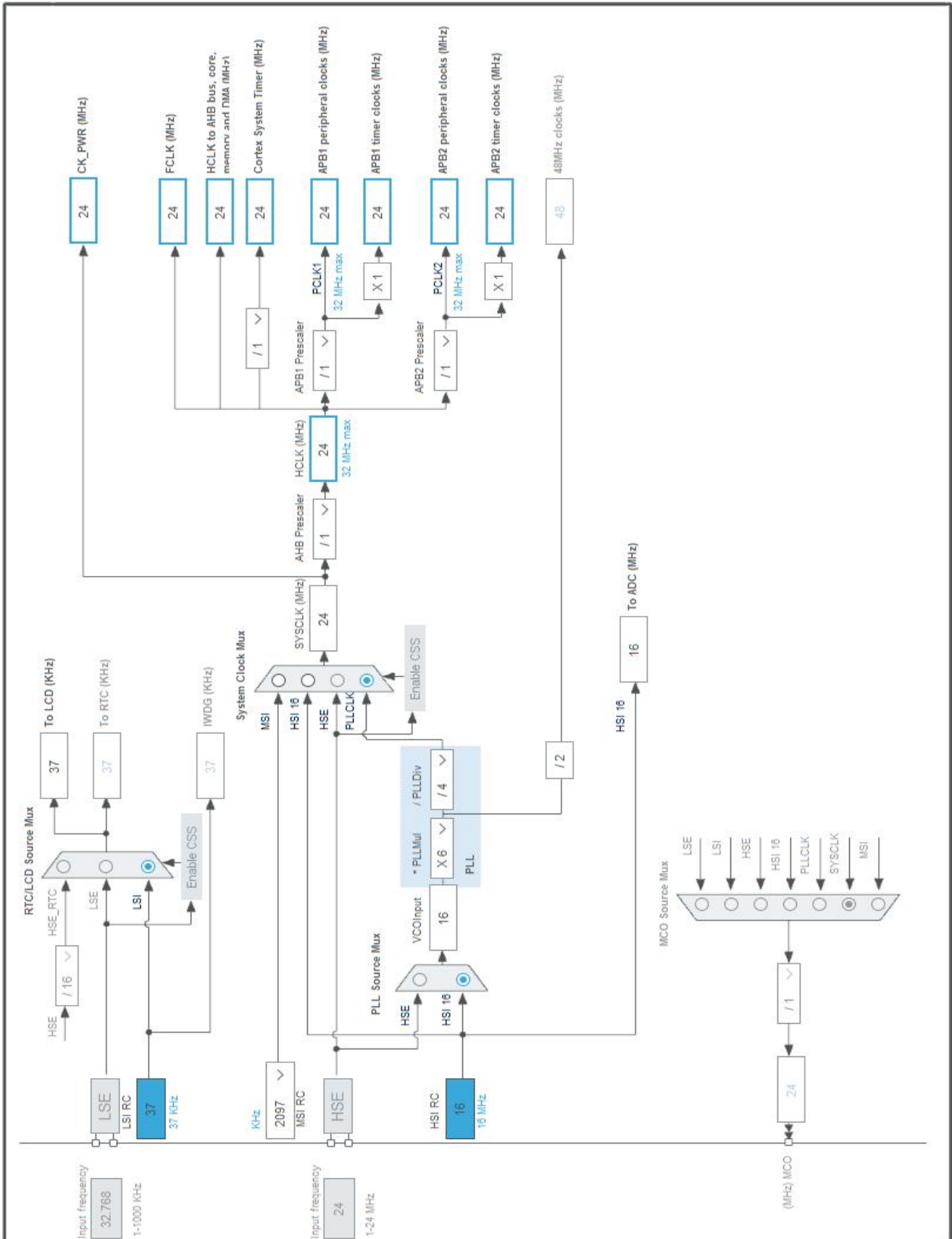
3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VLCD	Power		
7	NRST	Reset		
8	PC0	I/O	LCD_SEG18	
9	PC1	I/O	LCD_SEG19	
10	PC2	I/O	LCD_SEG20	
11	PC3	I/O	LCD_SEG21	
12	VSSA	Power		
13	VDDA	Power		
14	PA0-WKUP1	I/O	TIM5_CH1	
15	PA1	I/O	LCD_SEG0	
16	PA2	I/O	LCD_SEG1	
17	PA3	I/O	LCD_SEG2	
18	VSS	Power		
19	VDD	Power		
21	PA5	I/O	ADC_IN5	
22	PA6	I/O	LCD_SEG3	
23	PA7	I/O	LCD_SEG4	
24	PC4	I/O	LCD_SEG22	
25	PC5	I/O	LCD_SEG23	
26	PB0	I/O	LCD_SEG5	
27	PB1	I/O	LCD_SEG6	
28	PB2 *	I/O	GPIO_Input	
29	PB10	I/O	LCD_SEG10	
30	PB11	I/O	LCD_SEG11	
31	VSS	Power		
32	VDD	Power		
33	PB12	I/O	LCD_SEG12	
34	PB13	I/O	LCD_SEG13	
35	PB14	I/O	LCD_SEG14	
36	PB15	I/O	LCD_SEG15	
37	PC6	I/O	LCD_SEG24	
38	PC7	I/O	LCD_SEG25	
39	PC8	I/O	LCD_SEG26	
40	PC9	I/O	LCD_SEG27	
41	PA8	I/O	LCD_COM0	
42	PA9	I/O	LCD_COM1	

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
43	PA10	I/O	LCD_COM2	
47	VSS	Power		
48	VDD	Power		
50	PA15	I/O	LCD_SEG17	
51	PC10	I/O	USART3_TX	
52	PC11	I/O	USART3_RX	
53	PC12	I/O	USART3_CK	
55	PB3	I/O	LCD_SEG7	
56	PB4	I/O	LCD_SEG8	
57	PB5	I/O	LCD_SEG9	
58	PB6	I/O	USART1_TX	
59	PB7	I/O	USART1_RX	
60	BOOT0	Boot		
61	PB8	I/O	LCD_SEG16	
62	PB9	I/O	LCD_COM3	
63	VSS	Power		
64	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



5. Software Project

5.1. Project Settings

Name	Value
Project Name	CUBEMX.PROJECT
Project Folder	F:\XUE \CUBEMX.PROJECT
Toolchain / IDE	MDK-ARM V5.27
Firmware Package Name and Version	STM32Cube FW_L1 V1.10.3
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

5.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

5.3. Advanced Settings - Generated Function Calls

Rank	Function Name	IP Instance Name
1	MX_GPIO_Init	GPIO
2	MX_DMA_Init	DMA
3	SystemClock_Config	RCC
4	MX_ADC_Init	ADC
5	MX_LCD_Init	LCD
6	MX_USART1_UART_Init	USART1
7	MX_TIM5_Init	TIM5
8	MX_USART3_Init	USART3

6. Power Consumption Calculator report

6.1. Microcontroller Selection

Series	STM32L1
Line	STM32L151/152
MCU	STM32L152RCTx
Datasheet	DS8890 Rev12

6.2. Parameter Selection

Temperature	25
Vdd	3.0

6.3. Battery Selection

Battery	Li-SOCL2(AAA700)
Capacity	700.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	10.0 mA
Max Pulse Current	30.0 mA
Cells in series	1
Cells in parallel	1

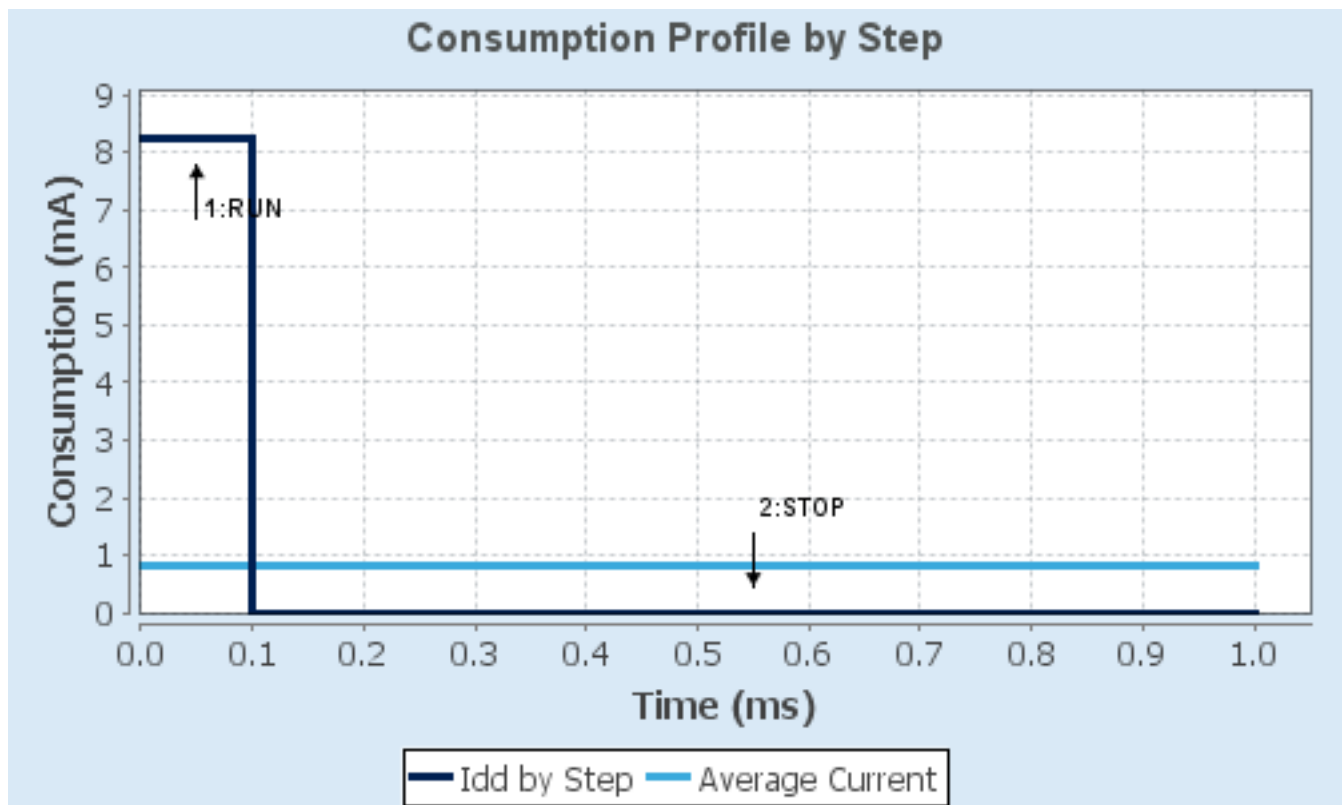
6.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-High	NoRange
Fetch Type	FLASH	n/a
CPU Frequency	32 MHz	0 Hz
Clock Configuration	HSEBYP PLL	ALL CLOCKS OFF
Clock Source Frequency	16 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	8.25 mA	435 nA
Duration	0.1 ms	0.9 ms
DMIPS	33.0	0.0
Ta Max	103.86	105
Category	In DS Table	In DS Table

6.5. Results

Sequence Time	1 ms	Average Current	825.39 μ A
Battery Life	1 month, 4 days, 21 hours	Average DMIPS	33.0 DMIPS

6.6. Chart



7. IPs and Middleware Configuration

7.1. ADC

mode: IN5

7.1.1. Parameter Settings:

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1
Resolution	ADC 8-bit resolution *
Data Alignment	Right alignment
Scan Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	End of sequence conversion
Low Power Auto Wait	Disabled
Low Power Auto Off	Disabled

ADC_Regular_ConversionMode:

Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
<u>Rank</u>	1
Channel	Channel 5
Sampling Time	4 Cycles

ADC_Injected_ConversionMode:

Number Of Conversions	0
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WatchDog:

Enable Analog WatchDog Mode	false
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7.2. GPIO

7.3. LCD

Mode: 1/4 Duty Cycle

mode: Multiplex Mode

mode: SEG0

mode: SEG1

mode: SEG2

mode: SEG3
mode: SEG4
mode: SEG5
mode: SEG6
mode: SEG7
mode: SEG8
mode: SEG9
mode: SEG10
mode: SEG11
mode: SEG12
mode: SEG13
mode: SEG14
mode: SEG15
mode: SEG16
mode: SEG17
mode: SEG18
mode: SEG19
mode: SEG20
mode: SEG21
mode: SEG22
mode: SEG23
mode: SEG24
mode: SEG25
mode: SEG26
mode: SEG27

7.3.1. Parameter Settings:

Clock Parameters:

Clock Prescaler	16 *
Clock Divider	16

Basic Parameters:

Duty Selection	1/4
Bias Selector	1/3 *
Multiplex mode	Enable

Advanced Parameters:

Voltage Source Selection	Internal
Contrast Control	

	2.73V *
Dead Time Duration	No dead Time
High Drive	Disable
Pulse ON Duration	0 pulse
Blink Mode	Disabled
Blink Frequency	fLCD/64 *

7.4. RCC

7.4.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	1 WS (2 CPU cycle)

RCC Parameters:

HSI Calibration Value	16
MSI Calibration Value	0
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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7.5. SYS

Timebase Source: SysTick

7.6. TIM5

Channel1: PWM Generation CH1

7.6.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Center Aligned mode1 *
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	Division by 2 *
auto-reload preload	

Enable *

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Enable *
CH Polarity	High

7.7. USART1

Mode: Asynchronous

7.7.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

7.8. USART3

Mode: Synchronous

7.8.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
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Clock Parameters:

Clock Polarity	Low
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Clock Phase
Clock Last Bit

One Edge
Disable

*** User modified value**

8. System Configuration

8.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC	PA5	ADC_IN5	Analog mode	No pull-up and no pull-down	n/a	
LCD	PC0	LCD_SEG18	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PC1	LCD_SEG19	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PC2	LCD_SEG20	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PC3	LCD_SEG21	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PA1	LCD_SEG0	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PA2	LCD_SEG1	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PA3	LCD_SEG2	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PA6	LCD_SEG3	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PA7	LCD_SEG4	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PC4	LCD_SEG22	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PC5	LCD_SEG23	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB0	LCD_SEG5	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB1	LCD_SEG6	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB10	LCD_SEG10	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB11	LCD_SEG11	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB12	LCD_SEG12	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB13	LCD_SEG13	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB14	LCD_SEG14	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB15	LCD_SEG15	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PC6	LCD_SEG24	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PC7	LCD_SEG25	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PC8	LCD_SEG26	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PC9	LCD_SEG27	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PA8	LCD_COM0	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PA9	LCD_COM1	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PA10	LCD_COM2	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PA15	LCD_SEG17	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB3	LCD_SEG7	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB4	LCD_SEG8	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB5	LCD_SEG9	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB8	LCD_SEG16	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
	PB9	LCD_COM3	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
TIM5	PA0-WKUP1	TIM5_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Very Low	
USART1	PB6	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	High *	
	PB7	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	High *	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
USART3	PC10	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	High *	
	PC11	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	High *	
	PC12	USART3_CK	Alternate Function Push Pull	No pull-up and no pull-down	High *	
GPIO	PB2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	

8.2. DMA configuration

DMA request	Stream	Direction	Priority
ADC	DMA1_Channel1	Peripheral To Memory	Low

ADC: DMA1_Channel1 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Half Word
Memory Data Width: Half Word

8.3. NVIC configuration

8.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	0	0
DMA1 channel1 global interrupt	true	0	0
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC global interrupt	unused		
LCD interrupt	unused		
USART1 global interrupt	unused		
USART3 global interrupt	unused		
TIM5 global interrupt	unused		

8.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	true	true	false
Hard fault interrupt	true	true	false
Memory management fault	true	true	false
Pre-fetch fault, memory access fault	true	true	false
Undefined instruction or illegal state	true	true	false
System service call via SWI instruction	true	true	false
Debug monitor	true	true	false
Pendable request for system service	true	true	false
System tick timer	true	true	true
DMA1 channel1 global interrupt	true	true	true

* User modified value

9. System Views

9.1. Category view

9.1.1. Current

10. Docs & Resources

Type	Link
Datasheet	http://www.st.com/resource/en/datasheet/DM00048356.pdf
Reference manual	http://www.st.com/resource/en/reference_manual/CD00240193.pdf
Programming manual	http://www.st.com/resource/en/programming_manual/CD00228163.pdf
Errata sheet	http://www.st.com/resource/en/errata_sheet/DM00105474.pdf
Application note	http://www.st.com/resource/en/application_note/CD00160362.pdf
Application note	http://www.st.com/resource/en/application_note/CD00167594.pdf
Application note	http://www.st.com/resource/en/application_note/CD00211314.pdf
Application note	http://www.st.com/resource/en/application_note/CD00259245.pdf
Application note	http://www.st.com/resource/en/application_note/CD00264342.pdf
Application note	http://www.st.com/resource/en/application_note/CD00264379.pdf
Application note	http://www.st.com/resource/en/application_note/CD00270081.pdf
Application note	http://www.st.com/resource/en/application_note/CD00273528.pdf
Application note	http://www.st.com/resource/en/application_note/CD00280599.pdf
Application note	http://www.st.com/resource/en/application_note/DM00024853.pdf
Application note	http://www.st.com/resource/en/application_note/DM00025071.pdf
Application note	http://www.st.com/resource/en/application_note/DM00032987.pdf
Application note	http://www.st.com/resource/en/application_note/DM00033344.pdf
Application note	http://www.st.com/resource/en/application_note/DM00040808.pdf
Application note	http://www.st.com/resource/en/application_note/DM00042534.pdf
Application note	http://www.st.com/resource/en/application_note/DM00073742.pdf
Application note	http://www.st.com/resource/en/application_note/DM00075930.pdf
Application note	http://www.st.com/resource/en/application_note/DM00085385.pdf
Application note	http://www.st.com/resource/en/application_note/DM00087593.pdf
Application note	http://www.st.com/resource/en/application_note/DM00129215.pdf
Application note	http://www.st.com/resource/en/application_note/DM00158601.pdf

Application note http://www.st.com/resource/en/application_note/DM00160482.pdf
Application note http://www.st.com/resource/en/application_note/DM00188260.pdf
Application note http://www.st.com/resource/en/application_note/DM00141025.pdf
Application note http://www.st.com/resource/en/application_note/DM00220769.pdf
Application note http://www.st.com/resource/en/application_note/DM00206898.pdf
Application note http://www.st.com/resource/en/application_note/DM00257177.pdf
Application note http://www.st.com/resource/en/application_note/DM00272912.pdf
Application note http://www.st.com/resource/en/application_note/DM00226326.pdf
Application note http://www.st.com/resource/en/application_note/DM00236305.pdf
Application note http://www.st.com/resource/en/application_note/DM00296349.pdf
Application note http://www.st.com/resource/en/application_note/DM00327191.pdf
Application note http://www.st.com/resource/en/application_note/DM00354244.pdf
Application note http://www.st.com/resource/en/application_note/DM00315319.pdf
Application note http://www.st.com/resource/en/application_note/DM00380469.pdf
Application note http://www.st.com/resource/en/application_note/DM00395696.pdf
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Application note http://www.st.com/resource/en/application_note/DM00660597.pdf